



eSim

SEMESTER LONG INTERNSHIP – AUTUMN 2026

INTERNSHIP SCREENING TASK REPORT

Task 7 – PCB Design using eSim

Enhanced Power Supply ESP32

Wide-Input (7–24 V) ESP32-C3 Development Board

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PCB – ESP32C3-EPS

1. Project Overview

This report presents the complete printed circuit board (PCB) design and schematic files for a custom, wide-input ESP32-C3 microcontroller development board. The design was developed as part of the screening task for the FOSSEE Autumn Internship 2026 program conducted by IIT Bombay.

The primary objective of this project was to design a fully functional, manufacturable development board centred around the ESP32-C3-WROOM-02 module — one that goes beyond a simple breakout board. Rather than exposing the module directly to an external supply, the board implements a complete power architecture that accepts a wide 7–24 V DC input, protects against reverse polarity and transient events, and steps the supply down through a two-stage regulator chain before it ever reaches the microcontroller.

The design was realised as a custom 4-layer PCB stack-up, optimised for power distribution, signal integrity, and RF-aware component placement around the module's integrated antenna.

Key Project Details

Microcontroller	ESP32-C3-WROOM-02 (RISC-V single-core, 160 MHz, Wi-Fi + BLE 5, integrated antenna)
Design Tools	FOSSEE eSim 2.5 (schematic capture) KiCad (PCB layout, integrated via eSim)
Power Input	Wide-range 7–24 V DC via dedicated DC input connector
Regulation Architecture	Protected input → LMR50410-Q1 buck (5 V) → LM1117-3.3 LDO (3.3 V)
USB Interface	USB Type-C, 16-pin — data/programming only, not the primary power source
PCB Stack-up	4-layer: F.Cu (power) / In1.Cu (signal) / In2.Cu (mixed) / B.Cu (ground)
Verification Status	Schematic connectivity and PCB design-rule checks completed with zero unresolved errors

The finished design includes a fully routed 4-layer stack-up and a 3D visual model for verifying component placement and mechanical fit ahead of fabrication.

2. Circuit Description

2.1 Power Supply Section

The board operates from a two-stage regulated supply rather than a single linear conversion from the wide input range. The external 7–24 V DC source first passes through an input protection stage comprising a TVS/Zener transient-clamping network and a P-channel MOSFET (STN3P6F6) configured for reverse-polarity protection. Under correct polarity the MOSFET conducts with very low resistance; if the supply is connected in reverse, it blocks the path entirely and protects every downstream component.

The protected input then feeds the LMR50410-Q1 synchronous buck converter, which efficiently steps the 7–24 V input down to an intermediate 5 V rail using a 4.7 uH inductor and a 10 uF local input capacitor placed close to the regulator to keep the high-frequency switching loop compact. The 5 V rail is not used directly by the MCU — it feeds an LM1117-3.3 LDO, which performs the final, low-noise regulation down to 3.3 V. Splitting the conversion this way lets the buck stage absorb the large voltage difference efficiently,

while the LDO concentrates purely on ripple rejection and supply cleanliness for the sensitive digital domain.

Local 4.7 uF bulk and 100 nF high-frequency decoupling capacitors are placed directly at the ESP32-C3 power pins to supply transient current locally and minimise supply impedance during rapid load changes (e.g. Wi-Fi transmit bursts).

2.2 Clock and Timing

A 32.768 kHz crystal is connected to the ESP32-C3's RTC circuitry to provide an accurate low-power timing reference for real-time clock and deep-sleep wake functions. The crystal is placed close to the module to minimise parasitic trace length and is routed away from the buck converter's switching node to avoid noise coupling into the oscillator.

2.3 Connectivity and Interfaces

- **USB Type-C Port:** A 16-pin USB-C receptacle provides the programming/debugging interface. Two 5.1 kΩ resistors on CC1/CC2 configure the board as a USB-C sink/device, and the D+/D− lines are protected by a dedicated USBLC6-2SC6 ESD device placed close to the connector.
- **GPIO Expansion:** Two 8-pin, 2.54 mm pitch headers (J3/J4) break out selected ESP32-C3 GPIOs for external sensors, actuators, displays, and breadboard prototyping.
- **Power Input:** A dedicated DC barrel/PCB-mount connector accepts the primary 7–24 V supply, kept electrically independent from the USB-C data interface.

2.4 User Interface

Two user-accessible push-button switches are provided: EN (Reset), pulled up through 10 kΩ and pulled low on press to reset the module, and IO9 (Boot), which changes the GPIO9 strap state during power-up/reset to enter the UART bootloader for firmware programming.

2.5 Peripheral Summary

Interface	Specification
Wireless	Wi-Fi 4 (802.11 b/g/n) + Bluetooth LE 5, integrated PCB antenna
USB	1 × USB Type-C (Full-Speed, programming/debug)
GPIO	2 × 8-pin, 2.54 mm expansion headers
Timing	32.768 kHz external RTC crystal
Control	EN (reset) and IO9 (boot) push buttons
Power Rails	5 V intermediate rail, 3.3 V logic rail

3. Schematic Screenshots

The complete schematic was captured in **FOSSEE eSim** and organised into clearly labelled functional blocks for readability and maintainability.

3.1 Full Schematic Overview

The schematic sheet below provides a top-level view of the entire circuit: the power-regulation chain (input protection → 5 V buck → 3.3 V LDO) across the top, USB-C & protection, boot/reset, and GPIO break-out blocks along the bottom, and the ESP32-C3 main controller block on the right — all interconnected through labelled nets.

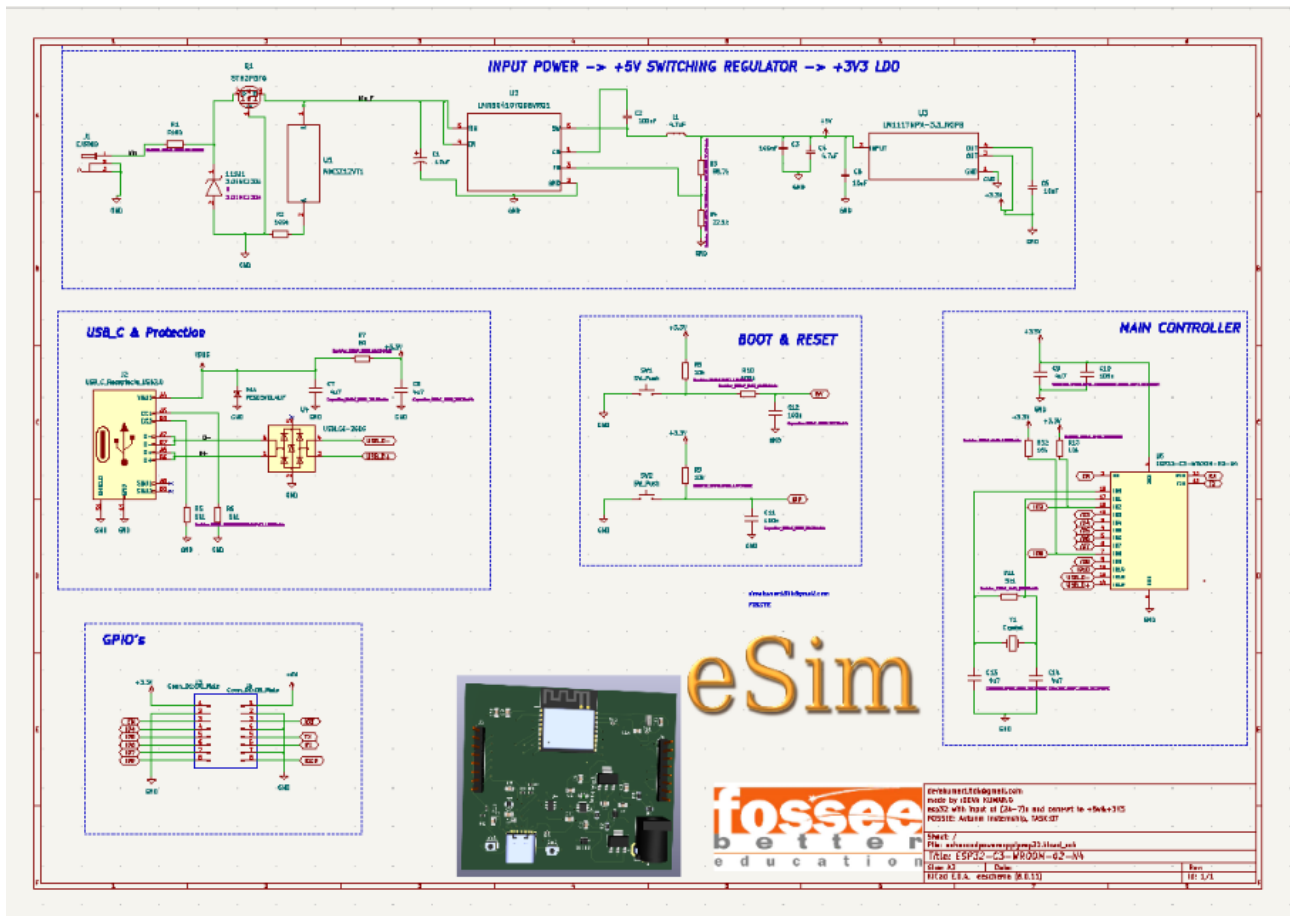


Figure 3.1 – Full schematic, captured in FOSSEE eSim

Note: Wire labels were used for power nets throughout the schematic (e.g. +3V3, +5V, GND) instead of floating global power symbols. This keeps the sheet visually clean while preserving correct electrical connectivity between blocks.

4. Footprint Assignment Details

Component footprints were assigned in the PCB workflow using standard eSim/KiCad libraries, cross-checked against each part's datasheet package outline to ensure correct pad geometry, courtyard clearance, and polarity marking before routing began.

4.1 Power Components

- **P-MOSFET (STN3P6F6):** SOT-23 package, reverse-polarity protection stage.
- **Buck Regulator (LMR50410-Q1):** SOT-23-6 package, with 4.7 uH shielded inductor footprint and associated input/output/bootstrap capacitor pads.
- **LDO (LM1117-3.3):** SOT-223 package, oriented for adequate copper pour around the thermal pad.

4.2 Passive Components

- **Resistors & Capacitors:** 0603 (1608 metric) SMD footprints — used for CC configuration resistors, pull-ups, and decoupling/bulk capacitors.
- **RTC Crystal:** 2-pad SMD crystal footprint with datasheet-specified load capacitors.

4.3 Connectors

- **USB Type-C:** 16-pin SMD receptacle footprint with mechanical anchor/ground tabs for board-edge structural support.
- **DC Power Input:** Through-hole barrel/PCB-mount connector footprint rated for the 7–24 V range.
- **GPIO Headers (J3/J4):** Standard 2.54 mm pitch, single-row, 8-way male header footprints.

4.4 User Interface

- **Tactile Switches (EN, IO9):** Standard 6×6 mm SMD tactile switch footprint.

5. PCB Layout

The PCB layout was completed using KiCad's PCB editor, accessed through the eSim integration. The board is implemented as a 4-layer stack-up, with functional grouping used to keep the ESP32-C3 module's RF section, the power-conversion cluster, and the user-accessible controls in clearly separated regions of the board.

5.1 Front Copper Layer (F.Cu) — Power / Component Routing

The front copper layer carries the primary power distribution and component-side connections: the protected input rail, the 5 V buck output, and the 3.3 V distribution to the ESP32-C3 and its decoupling network. Power components are grouped tightly in the central/lower region to keep high-current paths short.

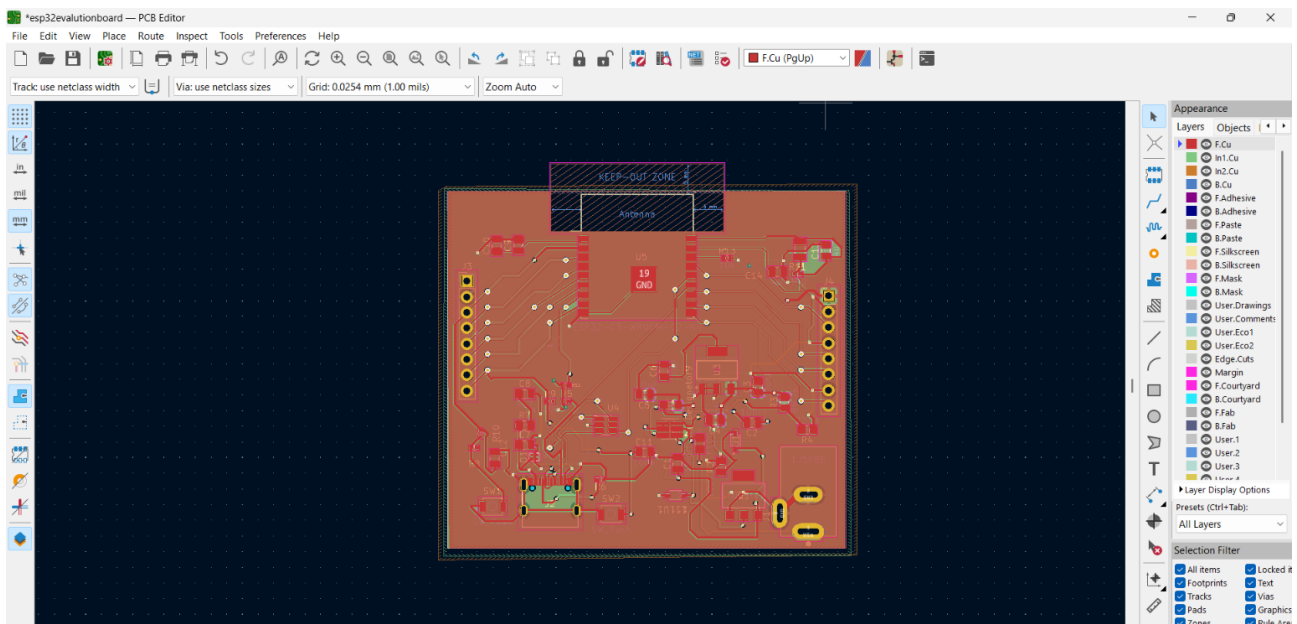


Figure 5.1 – F.Cu (top/power layer), showing the antenna keep-out zone above the module

5.2 Inner Layer 1 (In1.Cu) — Signal Routing

The first internal copper layer is dedicated to signal routing. The ESP32-C3 sits centrally with GPIO connections fanning outward toward the J3/J4 headers, reducing congestion on the outer copper layer.

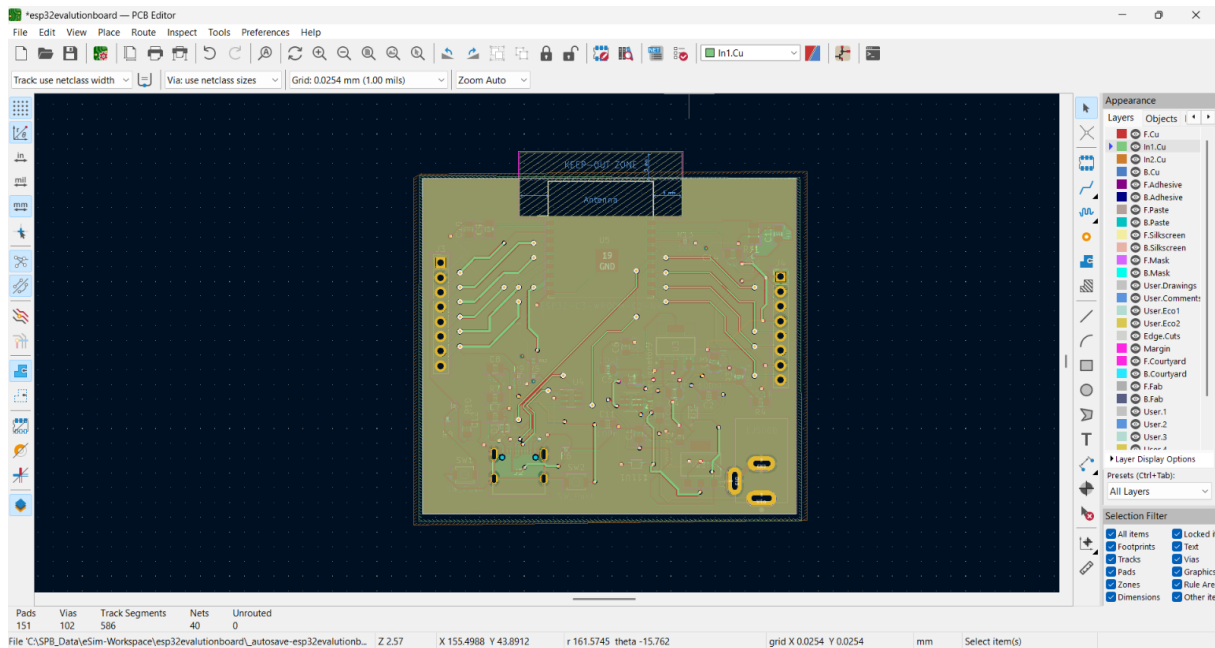


Figure 5.2 – In1.Cu, GPIO fan-out from the ESP32-C3 to the expansion headers

5.3 Inner Layer 2 (In2.Cu) — Mixed Routing / Distribution

The second internal layer provides additional routing channels for both signal and power nets where the outer layers become congested, preventing excessive routing density on the front and back copper.

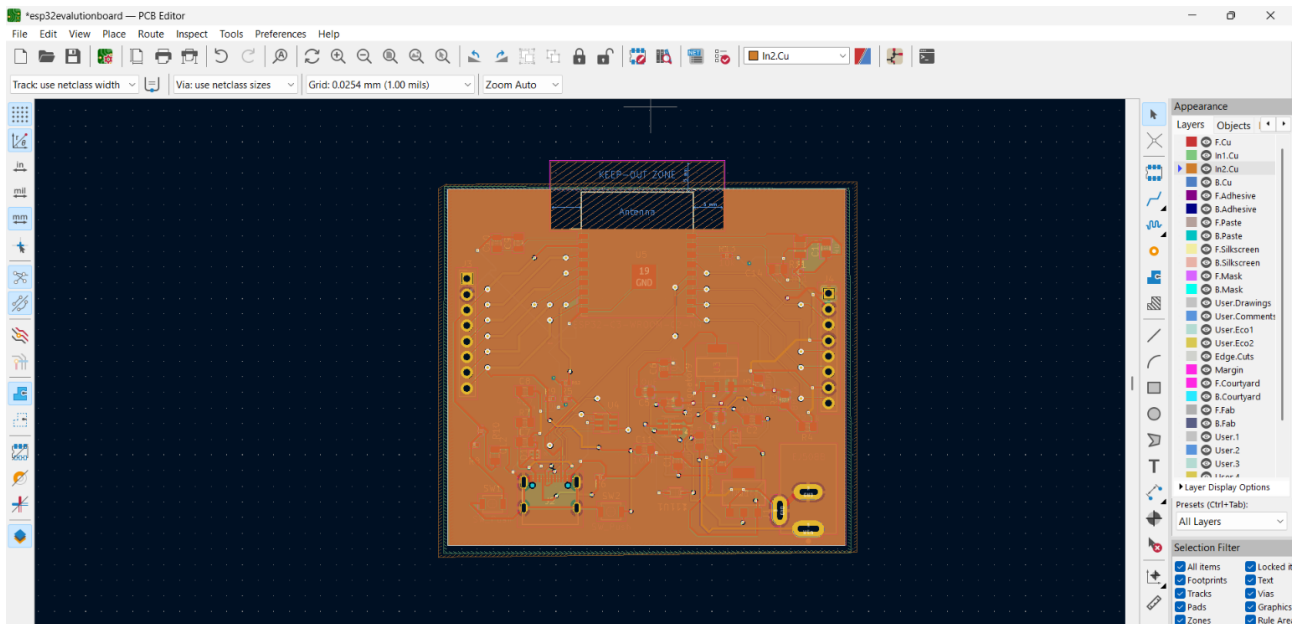


Figure 5.3 – In2.Cu, secondary power/signal distribution channel

5.4 Back Copper Layer (B.Cu) — Ground Plane

The back copper layer is dedicated almost entirely to a continuous ground plane, giving every signal a low-impedance return path directly beneath it. This is especially important for the ESP32-C3's integrated radio, which is sensitive to its surrounding electromagnetic environment.

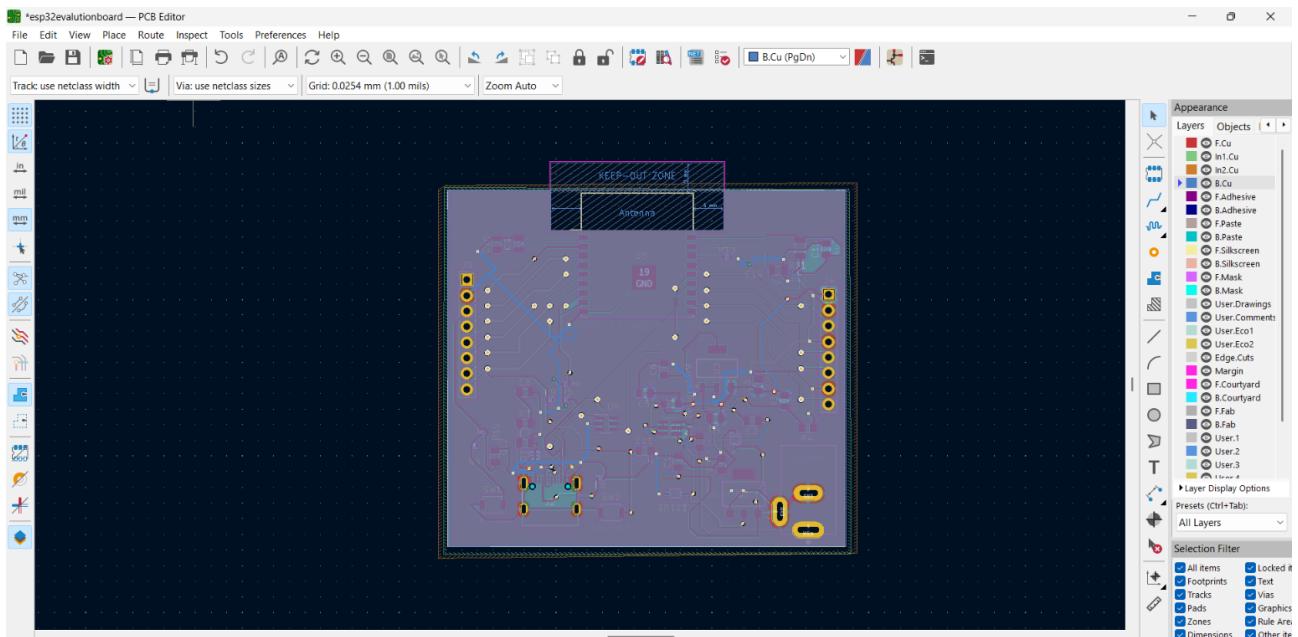


Figure 5.4 – B.Cu, continuous ground reference plane

Antenna Keep-Out Zone: A clearly marked keep-out region is enforced above the ESP32-C3 module on every copper layer (visible as the hatched “KEEP-OUT ZONE / Antenna” area). No copper, tracks, or vias are routed through this region, in line with the module manufacturer’s RF layout guidance, to prevent detuning or shielding the integrated antenna.

6. Routing Methodology

6.1 Layer Stack-up Strategy

The 4-layer stack-up was chosen specifically for a mixed power/RF/digital design of this kind:

Layer	Name	Function	Purpose
1	F.Cu (Front)	Power + component routing	Primary power distribution, short high-current paths
2	In1.Cu (Inner 1)	Signal routing	GPIO fan-out from ESP32-C3 to expansion headers
3	In2.Cu (Inner 2)	Mixed routing / distribution	Secondary power & signal routing channel
4	B.Cu (Back)	Ground plane	Continuous low-impedance return path

This arrangement ensures the ground plane on B.Cu sits close to the routing layers above it, minimising loop area and unwanted electromagnetic coupling — particularly important given the ESP32-C3’s onboard radio.

6.2 Signal & Power Routing Guidelines

- High-current power paths (DC input, MOSFET, buck converter, inductor, 5 V rail, LDO) are kept short and comparatively wide.
- Switching node of the buck converter is kept as small and compact as practical to limit EMI coupling into adjacent traces.

- Decoupling capacitors are placed as close as physically possible to the ESP32-C3 power pins.
- USB D+/D– are routed as a matched differential pair, with the ESD protection device placed close to the connector to keep the unprotected trace length short.
- RTC crystal traces are kept short and isolated from the switching regulator's noisy node.
- Ground plane continuity: no signal traces cross splits or gaps in the ground plane, avoiding return-path discontinuities.

6.3 Design Rule Constraints

All routing was carried out within standard 4-layer PCB fabrication constraints:

- Minimum trace width and spacing within standard manufacturer capabilities.
- Minimum via drill size and annular ring within standard fabrication tolerances.
- Copper-to-edge clearance maintained around the board outline.
- No acid traps or acute-angle traces in the routing.

7. Design Verification

Before committing the design to fabrication, both schematic-level and PCB-level checks were carried out in eSim/KiCad.

7.1 Schematic Checks (ERC)

- Every power net verified (protected input, 5 V, 3.3 V, GND).
- ESP32-C3 power pins and decoupling network checked.
- EN/reset and IO9/boot strapping circuitry verified.
- USB D+/D– connections and CC resistor values verified.
- Checked for unintended VBUS-to-3.3 V connections.

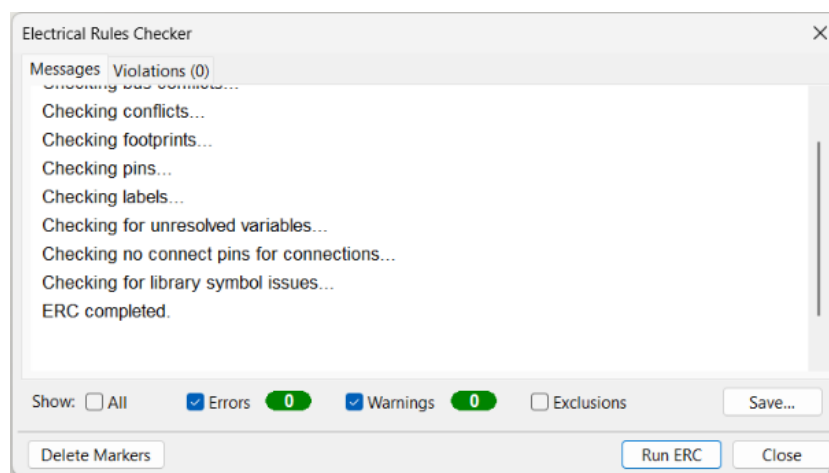


Figure 7.1 – Electrical Rules Checker: 0 errors, 0 warnings on ERC completion

7.2 PCB Checks (DRC)

- Design Rule Check run with zero unresolved clearance/track-width violations.
- Via placement, pad sizing, and copper-zone connections verified.

- Antenna keep-out region checked against the module manufacturer's guidance.
- Silkscreen overlaps, connector orientation, and footprint pin numbering verified.
- Board outline and mounting/mechanical dimensions verified.

Result: the schematic and PCB layout passed the above checks with all nets fully routed and no outstanding design-rule violations, indicating the board is ready for Gerber export and fabrication.

8. 3D Model

A 3D render of the board was generated from the routed PCB layout to visually verify component placement, connector accessibility, and overall mechanical form factor ahead of fabrication.

8.1 Isometric View

The render below shows the ESP32-C3-WROOM-02 module at the top of the board with a clear area around its antenna, the power-conversion cluster and USB-C/DC input in the centre, boot/reset buttons along the bottom edge, and the two GPIO expansion headers along the left and right sides.

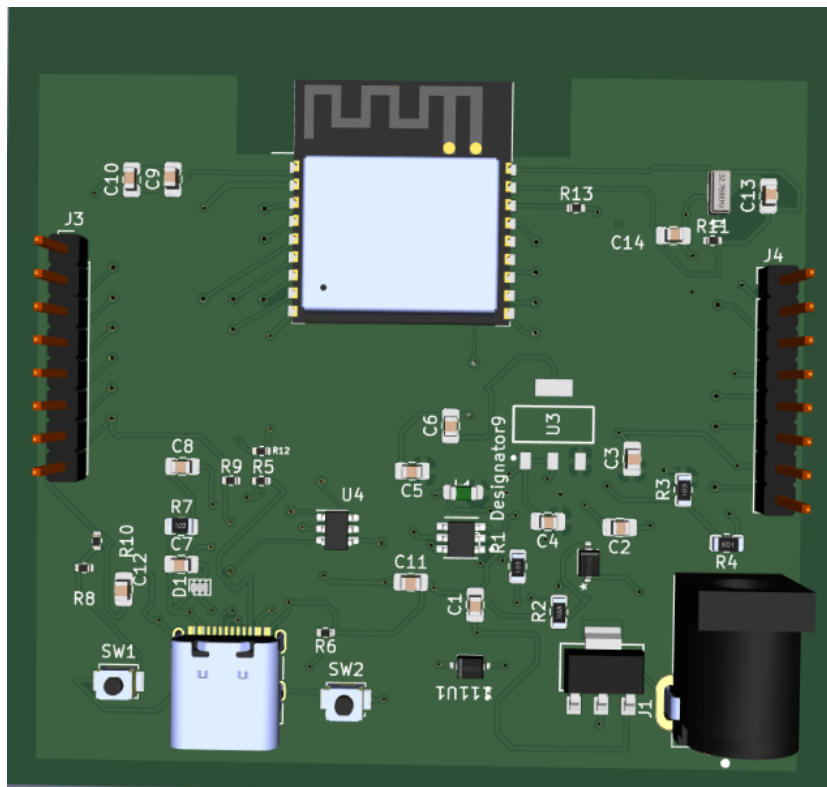


Figure 8.1 – 3D render of the fully routed board

9. Design Considerations and Conclusion

9.1 Design Considerations

Power Integrity: The two-stage buck + LDO architecture means the LDO never has to absorb the full 7–24 V range — it handles only the 5 V→3.3 V step, giving a clean, low-impedance supply to the ESP32-C3 even under wide input variation.

Signal Integrity: The 4-layer stack-up gives every routed signal an adjacent reference plane, minimising return-current loop area and reducing coupling into the RTC crystal and USB differential pair.

RF Considerations: A dedicated antenna keep-out zone above the ESP32-C3 module, free of copper, tracks, and vias, protects the integrated antenna's radiation performance.

Protection: Reverse-polarity protection (P-MOSFET) and transient protection (TVS/Zener) on the input, plus dedicated USB ESD protection, guard the board against the most common field-failure modes.

Manufacturability: The design was routed within standard 4-layer fabrication constraints — standard trace/space, via drill sizes, and copper-to-edge clearances — avoiding the need for premium manufacturing processes.

Expandability: Two 2.54 mm GPIO headers turn the board from a closed embedded controller into a reusable development platform for sensors, displays, and other peripherals.

9.2 Conclusion

This project delivers a complete, manufacturable PCB design for a wide-input ESP32-C3 development board. The design achieves the following key outcomes:

- **Verified design** — schematic connectivity and PCB design-rule checks completed with no outstanding errors.
- **Robust wide-input power architecture** — a 7–24 V input is protected, converted, and regulated through a two-stage buck + LDO chain before reaching the MCU.
- **Professional-grade 4-layer stack-up** — dedicated signal, distribution, and ground layers support both power integrity and RF performance.
- **Practical connectivity** — USB-C programming/debug, RTC timing, hardware boot/reset control, and GPIO expansion make the board a genuinely reusable development platform.
- **Complete documentation** — schematic, layer-by-layer PCB views, routing methodology, verification checks, and a 3D render are all included for full design transparency.

Overall, the project demonstrates the complete hardware-development workflow — from system architecture and component selection, through schematic design and protection circuitry, to multilayer PCB layout, RF-aware placement, and design verification — using the FOSSEE eSim / KiCad-based design environment. The resulting board is not merely an ESP32 carrier board; it is a complete embedded hardware platform combining wide-range power management, protection, wireless processing, USB connectivity, hardware boot control, RTC support, and expandable I/O in a single four-layer PCB design.

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